

Force-Directed Algorithm

The implemented algorithm is based on **Eades' model**, a spring-embedder variation that treats the graph as a physical system of particles. Equilibrium is reached by iteratively calculating two opposing forces for every node v .

1. Repulsive Force (Coulomb's Law)

Every pair of nodes (u, v) exerts a repulsive force to prevent overlap and distribute nodes across the canvas. The magnitude of this force F_r is inversely proportional to the square of the distance between them:

$$F_r(u, v) = \frac{k_r}{d^2}$$

Where:

- k_r : The Repulsion constant (defined in the UI).
- d : The Euclidean distance $\sqrt{(x_v - x_u)^2 + (y_v - y_u)^2}$.

2. Attractive Force (Hooke's Law)

Nodes connected by an edge $(u, v) \in E$ are pulled together by an attractive force F_a , proportional to their distance:

$$F_a(u, v) = k_a \cdot d$$

Where:

- k_a : The Attraction constant (defined in the UI).
- d : The current distance between the connected nodes.

3. Net Force Calculation

For any node v , the net force vector $\vec{F}_{net}$ is the sum of all repulsive forces from every other node and all attractive forces from its direct neighbors:

$$\vec{F}_{net}(v) = \sum_{u \neq v} \vec{F}_r(u, v) + \sum_{(u, v) \in E} \vec{F}_a(u, v)$$

4. Displacement and Damping

In each iteration, the position P of node v is updated using a **Damping factor** (η) to simulate friction and prevent infinite oscillation:

$$P_{new} = P_{old} + (\vec{F}_{net} \cdot \eta)$$

The system iteratively approaches a state where $\vec{F}_{net} \approx 0$, resulting in a stable, aesthetically balanced layout.